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How to Understand Your Sleep Study Results

A Practical, Evidence-Based Framework to Decode Your Sleep Health

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MEDICAL DISCLAIMER

This guide is intended for informational and educational purposes only and is not a substitute for professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider before starting, changing, or discontinuing any supplement, medication, or health regimen.

Some treatments or devices discussed may not be FDA-approved for all sleep disorders. Sleep studies should always be interpreted by a qualified physician. This guide is intended to help you understand your results, not self-diagnose.

Individual responses vary, and dosages listed here reflect ranges studied in scientific literature, not personalized medical recommendations.

ABOUT THE AUTHOR

Hello there!

I'm Sahil Chopra, MD. I'm a quadruple-boarded physician specializing in Internal, Sleep, and Pulmonary/Critical Care Medicine. I completed my Internal Medicine residency at UCLA, Pulmonary/Critical Care fellowship at Loma Linda University, and Sleep Medicine training at Harvard.

We started Empower Sleep with one vision in mind: to provide Harvard-grade sleep care to as many patients as possible. The strategies I adopted in my Sleep Medicine fellowship were deeply rooted in the physiology of sleep science. We now apply this science here at Empower Sleep. The goal is to treat all patients as an "n-of-1" (individualized approach). Not a homogeneous, population health-based approach, which is what happens in traditional healthcare. Everyone's sleep is unique to them, as is their sleep disorder(s). When we approach every patient as an individual, we can typically get to the root cause, find a personalized solution, and have stellar clinical outcomes.

At Empower Sleep, we have cared for thousands of patients and analyzed hundreds of thousands of nights of sleep biomarkers. Our innovative care model and success is a by-product of my mentors, my amazing team, our patients, and my family.

Thank you in advance for taking the time to improve your sleep health. I hope you find this guide as a helpful tool to solve your sleep struggles.

It is my wish for all humans to live healthy and fulfilled lives.

Sahil Chopra, MD

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Introduction

As a sleep physician, one of the most common things I hear from patients after a sleep study is: *"Doctor, I have no idea what these numbers mean."* This feeling is completely normal. I used to feel the same way even as an internal medicine resident. It wasn't until my Sleep Medicine Fellowship at Harvard that I was able to understand and explain the significance of these metrics.

Sleep studies are filled with technical jargon, graphs, and percentages that can feel overwhelming. Yet hidden in those pages are answers to some of the most important questions about your health—why you feel exhausted, why your blood pressure is hard to control, or why your mood and focus aren't what they used to be.

My goal with this guide is to help you interpret those results in a clear, practical way. Think of this as a mini guide. By the end, you should feel confident enough to look at your sleep study and understand what it tells you about your sleep, your health, and the next steps you can take to feel better.

A “Sleep Study” is a tool to measure the health of your sleep.

Sleep testing is a tool that measures what is happening to your physiology during the window of sleep. **Please don't think of it as just a “sleep apnea test**.”

Sleep testing has evolved dramatically over the last few decades. Today, we have several different tools, ranging from the full in-lab polysomnogram to home-based devices that track airflow and oxygen.

Let me give you an overview of some of the different systems. All of these are FDA-approved.

Polysomnography (PSG) – The Gold Standard

This is the overnight test performed in a sleep lab. Electrodes measure your brain activity (EEG), eye movements (EOG), muscle activity (EMG), breathing patterns, oxygen levels, heart rhythm, and leg movements. PSG provides the most comprehensive, accurate and precise picture of sleep [1]. The drawback is having to sleep in a lab and cost: it can be anywhere between $1,000-$8,000, depending on the facility.

Flow-Based Studies, Home Sleep Apnea Testing (HSAT)

This is a simplified version of PSG done at home. It typically includes airflow sensors, oxygen saturation, chest and abdominal belts, and sometimes, a position sensor. It is highly effective for diagnosing obstructive sleep apnea but may not capture sleep stages or limb movements [2].

Cardiopulmonary Coupling (CPC) with SleepImage

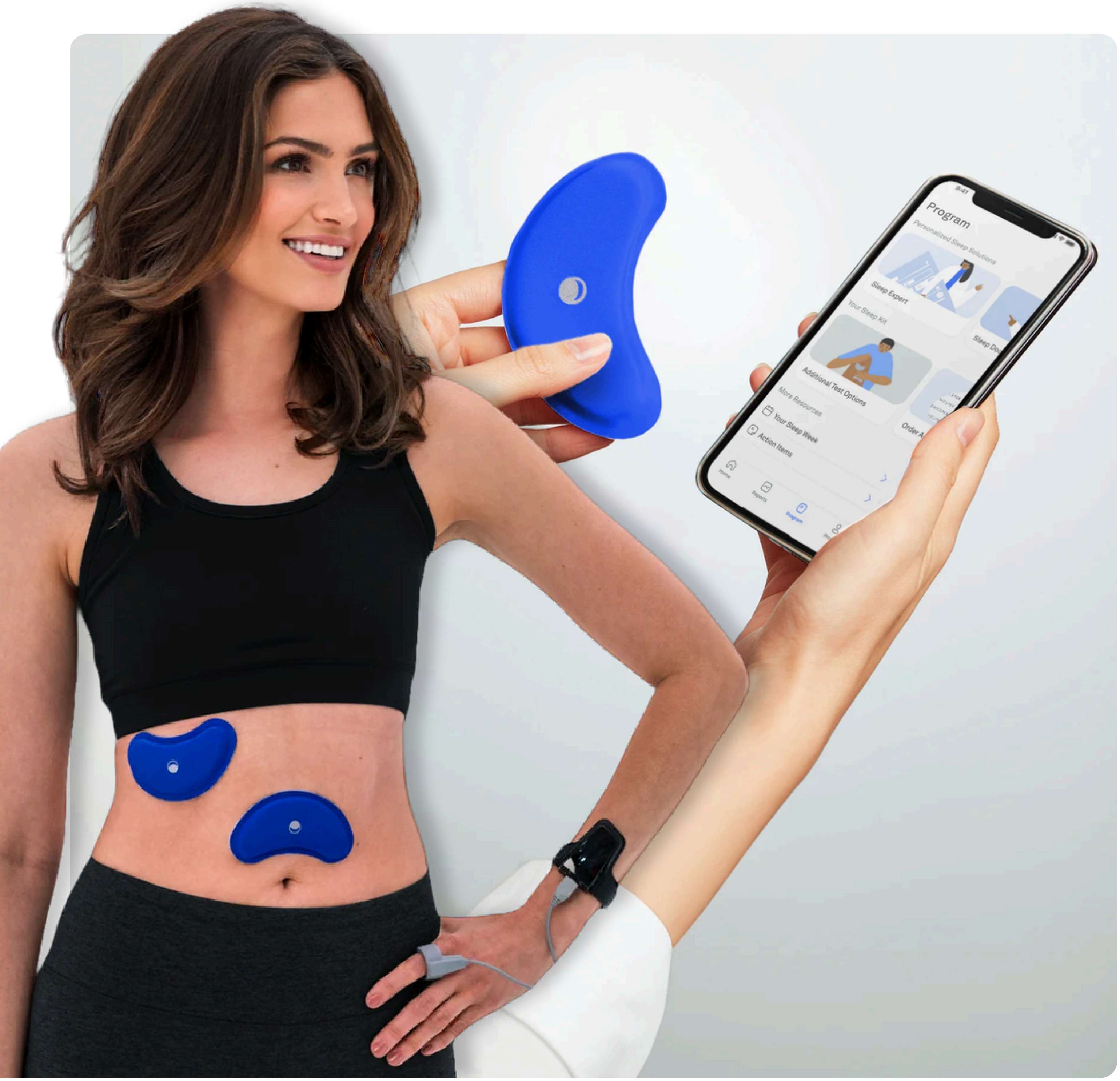
CPC uses a ring-based oximeter and photoplethysmography (PPG), the red or green light inside most wearables, to measure heart rate, oxygenation, and respiration to understand the interaction of breathing and heart rate. It can shed light on your sleep health, breathing, oxygen, heart rate, movement and more [3].

Peripheral Arterial Tonometry (PAT)

PAT devices (such as the WatchPAT) measure changes in vascular tone, body position, oxygen levels, and movement.

Wesper Sleep Testing System

Wesper is a home sleep testing system that uses thin, wireless patches placed on the chest and abdomen. Wesper's sensors measure breathing effort (thoracic and abdominal motion), snoring, airflow patterns, body position, oxygen saturation, sleep time, and stages.



Onera Sleep Testing System

Onera is a **medical-grade, patch-based sleep testing platform** designed to make polysomnography more accessible outside the lab. Instead of dozens of wires, Onera uses **wireless patches** placed on the chest, forehead, and legs to capture many of the same signals as an in-lab study. We are able to measure EEG (brain waves), EOG (eye movement), EMG (muscle tone), breathing effort, airflow, leg movement, body position and heart rate.

Strengths:

- **PSG-level data** in a home environment
- Wireless, fast setup (5–10 minutes)
- FDA-cleared, validated against in-lab PSG [27]
- Provides full **sleep staging** (unlike most HSAT devices)

A Comparison of Sleep Tests

Tool	What It Measures	Strengths	Limitations	Use Case
Polysomnography	Brain waves, breathing, O_2, heart rhythm, leg movements	Gold standard, comprehensive	Requires lab, higher cost	Complex cases, diagnosis beyond OSA
HSAT	Airflow, O_2, breathing effort	Convenient, home-based, cost-effective	Limited data (no sleep staging, limb movements)	Diagnostic only
Cardiopulmonary Coupling (CPC) with SleepImage	Photoplethysmography (PPG), Oximetry	Multi-night tracking, captures autonomic and sleep health	Indirect	Both diagnosis, tracking and optimization
PAT	Arterial tone, O_2, actigraphy	Detects arousals, REM-related OSA	Staging less precise than PSG	Single-night diagnostic
Onera	EEG, EOG, EMG, breathing effort, O_2, leg movements	Near-PSG data at home, wireless, FDA-cleared	Cost, newer tech	Home PSG replacement
Wesper	Breathing effort, airflow, O_2, position, snoring, staging algorithms	Comfortable, wireless, multi-night, FDA-cleared	No EEG, requires smartphone	Tech-forward home OSA testing

01 Key Metrics

Apnea-Hypopnea Index (AHI)

The AHI counts how many times per hour your breathing either completely stops (apnea) or is partially impaired (hypopnea) [5]. The *number of events per hour determines the severity of sleep apnea:*

For Pediatrics	For Adults
Normal: 0	**Normal:** 0-5
Mild OSA: 1-5 events/hour	**Mild OSA:** 5-14 events/hour
Moderate OSA: 5-10 events/hour	**Moderate OSA:** 15-29 events/hour
Severe OSA: ≥ 10 events/hour	**Severe OSA:** ≥ 30 events/hour

Oxygen Saturation (SpO_2)

Reports will highlight the lowest oxygen level reached (nadir) and how long you spent below 90% or 88%. Frequent drops in oxygen can stress the heart, brain, and other organs [6]. The lower the O2, the worse the sleep apnea and the higher the probability of downstream health issues. If you live at altitude, we'll look at this with more detail.

Arousal Index and Sleep Fragmentation

This number reflects how often your sleep is disrupted by arousals (brief awakenings, sometimes only a few milliseconds long). Even if you don't remember them, frequent arousals lead to poor-quality sleep [7]. The more the arousals, the worse your sleep health.

Periodic Limb Movement (PLM) Index

This number reflects how many limb movements, particularly of your lower extremities, you have. These may or may not be disruptive to your sleep. The higher the number of limb movements, the more disruptive your sleep. Sometimes this can give a clue to restless leg syndrome.

Sleep Efficiency (SE)

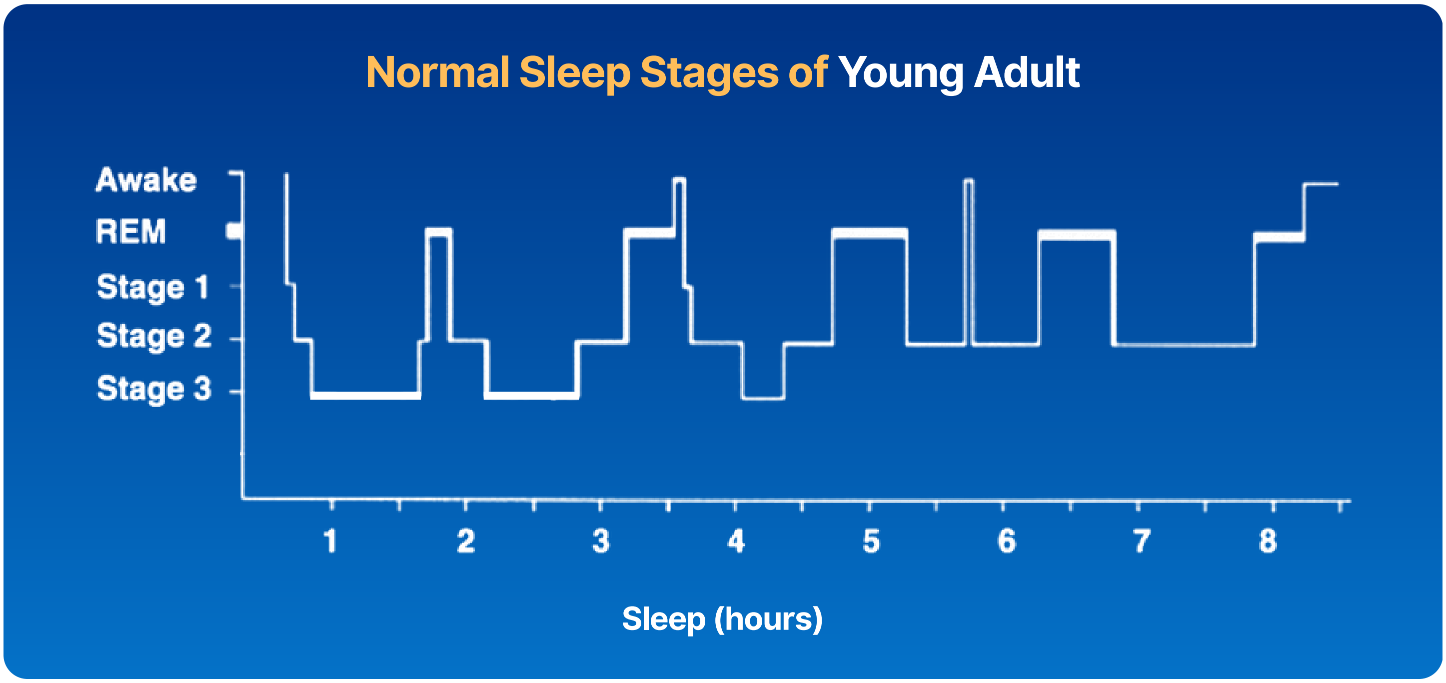
The percentage of time spent asleep relative to total time in bed. A healthy efficiency is usually >85% [8].

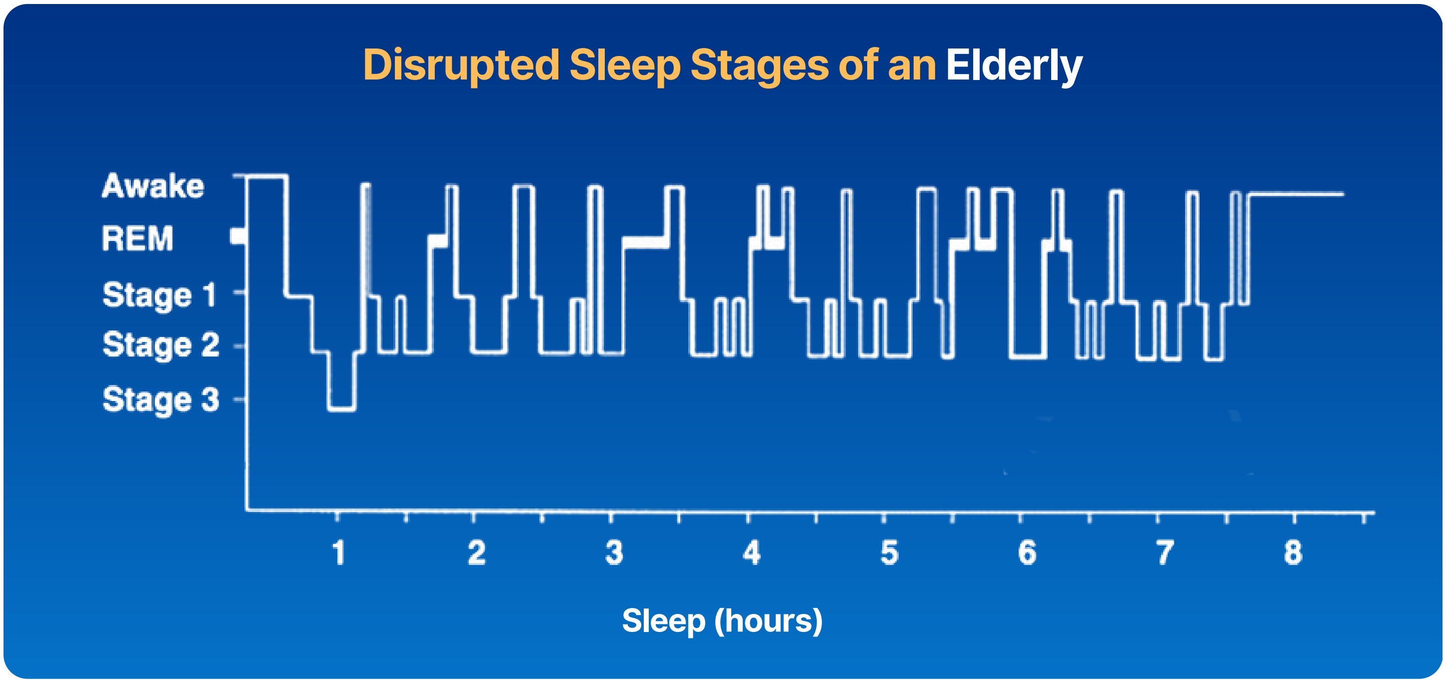
Hypnogram & Sleep Architecture

Your report may include a hypnogram—a graph showing sleep stages across the night.

- **N1 (light sleep) → N2 (transitional sleep) → N3 (deep sleep/slow-wave sleep) → REM (dream sleep)**. I have a separate booklet on the importance on the different stages of sleep. Email us at care@empowersleep.com for more information.

- A healthy night cycles through these about every 90 - 120 minutes.
- **Normal sleep stages**: cycling smoothly, with ~20–25% REM, 15–20% deep sleep, 50-65% light sleep.
- **Disrupted sleep stages**: fragmented, reduced deep sleep, shortened REM, or multiple awakenings (common in sleep apnea, insomnia, and PLMD) [9].





02 Understanding AHI 3% vs. AHI 4%

When you read your sleep study, you may see two different apnea-hypopnea indices: AHI 3% and AHI 4%. At first glance, this seems like a minor technical detail, but it can make a big difference in how your results are interpreted.

The Basics

- **Apnea,** where breathing completely stops for ≥10 seconds, is always counted the same way.
- **Hypopnea**, in other words a partial reduction in airflow, is where the difference occurs.
    - **AHI 3% rule**: A hypopnea is scored if airflow drops by at least 30% for ≥10 seconds and is accompanied by either a micro-awakening (arousal) or a ≥3% drop in oxygen.
    - **AHI 4% rule**: A hypopnea is scored if airflow drops by at least 30% for ≥10 seconds and is accompanied by a ≥4% drop in oxygen (arousals don't count).

The History

- The **American Academy of Sleep Medicine (AASM)** originally allowed either the 3% or 4% definition.
- **Medicare and many insurance companies** standardized on the 4% rule, largely to limit over-diagnosis of sleep apnea and possibly limiting patients qualifying for CPAP therapy.
- **Clinicians and researchers** often prefer the 3% rule, since it captures more subtle breathing disturbances linked to symptoms like fatigue, insomnia, and cardiovascular risk.
- As a result, two patients with the exact same study might be labeled "mild OSA" under the 3% criteria but "no OSA" under the stricter 4% criteria.

Why This Matters for You

- **3% AHI is more sensitive** → picks up breathing disturbances that fragment sleep, even without big oxygen drops.
- **4% AHI is more conservative** → may miss clinically significant cases, especially in women, younger patients, or those with insomnia.

- If your doctor says “your study was negative,” it’s worth asking whether they used the 3% or 4% criteria.

Key Takeaway

Neither definition is “wrong” — they’re just different thresholds. At Empower Sleep, we look at both and interpret them in the context of your symptoms, health risks, and daytime functioning.

The goal is not just to chase numbers but to restore your energy, health, and quality of life.

My personal preference is to use the 3% threshold so I can get patients “qualified” for a treatment if its clinically indicated.

03 AHI vs. RDI, what’s the difference?

When you look at your sleep study, you may see two different numbers: the Apnea-Hypopnea Index (AHI) and the Respiratory Disturbance Index (RDI). They sound similar, but they are not identical. Understanding the difference is important.

Apnea-Hypopnea Index (AHI)

- Counts the number of apneas (complete pauses in breathing) and hypopneas (partial reductions in breathing) per hour of sleep.
- Depending on whether the 3% or 4% rule is used, the hypopneas included may vary.
- This is the most commonly cited number in sleep medicine and the one typically used to define mild, moderate, or severe sleep apnea.

Respiratory Disturbance Index (RDI)

- Includes all apneas + hypopneas, plus another category: respiratory effort-related arousals (RERAs).
- RERAs are events where breathing becomes labored, airflow is restricted, and the brain briefly wakes up (arousal), but oxygen levels don’t necessarily drop enough to meet hypopnea criteria.

- In other words:
  - AHI = apneas + hypopneas
  - RDI = apneas + hypopneas + RERAs

Why This Matters

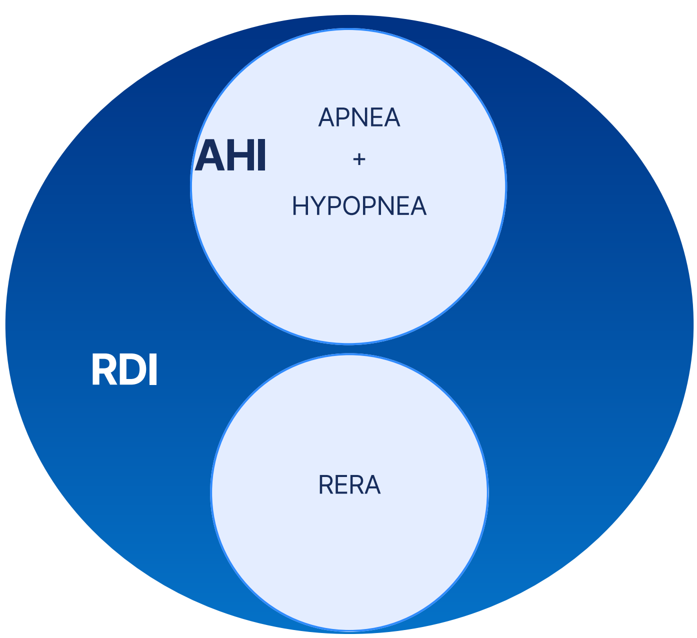
- Some patients have a low AHI but a high RDI — meaning they don't technically meet criteria for sleep apnea, yet their sleep is still fragmented by repeated breathing-related arousals.
- This condition is sometimes called Upper Airway Resistance Syndrome (UARS).
- Patients with UARS often complain of fatigue, insomnia, unrefreshing sleep, headaches, and difficulty concentrating, despite being told their "AHI is normal."

Insurance vs. Clinical Reality

- Insurance carriers often use AHI (4% rule) as the benchmark for CPAP coverage.
- Clinically, however, a high RDI can be just as significant as a high AHI, especially if you're symptomatic.

Key Takeaway

If your sleep study shows a low AHI but a high RDI, it means your breathing is still disrupting your sleep — even if oxygen levels look "okay." At Empower Sleep, we always look at both indices together and treat the person, not just the number.



04 PUTTING IT ALL TOGETHER — CASE EXAMPLES

- **Mild OSA:** AHI3% 8/hr, O_2 nadir 89%, sleep efficiency 82%. Fatigue, snoring, morning headaches.
- **Severe OSA:** AHI 48, O_2 nadir 72%, fragmented hypnogram. Associated with uncontrolled hypertension.
- **Severe PLMD:** PLM index >50/hour, arousal index >25, efficiency 60%. Complaints of restless legs and unrefreshing sleep [14].
- **Insomnia:** AHI 2, RDI 5. Sleep onset latency 50 minutes, total sleep time 4.5 hours. Sleep efficiency of 50%. Hypnogram shows prolonged awakenings and low REM percentage.

05 WHAT YOUR RESULTS MEAN FOR YOUR HEALTH

1. Heart & Circulatory Health

- Sleep apnea is strongly associated with hypertension, atrial fibrillation, and heart failure [15–17].
- Even mild untreated apnea can blunt the normal nighttime blood pressure “dip.”

2. Brain & Cognitive Function

- Repeated arousals impair deep sleep and therefore influence attention and memory [18].
- Untreated OSA increases dementia risk as well as an increased risk of developing a stroke [19].
- Patients often report brain fog and mood swings tied to fragmented hypnograms.

3. Metabolic & Hormonal Health

- OSA increases insulin resistance and diabetes risk [20].
- Sleep fragmentation elevates cortisol [21].
- Apnea in men lowers testosterone [22].

4. Mood, Mental Health & Energy

- PLMD and insomnia can cause fatigue and depression despite “normal AHI” [23].
- Reduced REM is linked to irritability and anxiety.

5. Longevity & Aging

- Severe untreated OSA is linked to higher all-cause mortality [24].
- Poor sleep accelerates biological aging [25].

06 NEXT STEPS

- **Therapies**: Behavioral therapy for insomnia, CPAP, oral appliances, positional therapy, surgery.
- **Lifestyle**: Weight management, alcohol reduction, nasal breathing, sleep hygiene.
- **Follow-up**: If you’ve “tried everything” and still don’t sleep well, schedule a complimentary consult with our team for guidance.

07 CONCLUSIONS

Understanding your sleep study is the first step toward reclaiming your health. Don’t get lost in the numbers — look at the whole picture. At Empower Sleep, we believe your treatment should be guided not only by indices but also by your lived experience, symptoms, and long-term health goals.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